

BINARY CLASSIFICATION USING LOGISTIC REGRESSION

Telco Customer Churn Prediction

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ABSTRACT

This project implements a binary classification model using Logistic Regression to predict customer churn in the telecommunications industry. Using the Telco Customer Churn dataset comprising 7,043 customers with 20 features, we developed a complete machine learning pipeline including exploratory data analysis, preprocessing, model building, and comprehensive evaluation. The final model achieved an accuracy of 80.1% with an ROC-AUC score of 0.856, demonstrating strong discriminative ability. Key findings reveal that contract type, internet service, and technical support are the primary drivers of customer churn. This report documents all six phases of the project with detailed methodology, results, visualizations, and recommendations for future improvements.

PHASE 1: EXPLORATORY DATA ANALYSIS

1.1 Problem Overview

Customer churn represents a significant challenge for telecommunications companies. Predicting which customers are likely to churn enables proactive retention strategies, reducing revenue loss and improving customer satisfaction.

1.2 Dataset Description

Dataset: Telco Customer Churn (Kaggle)

Total Records: 7,043 customers

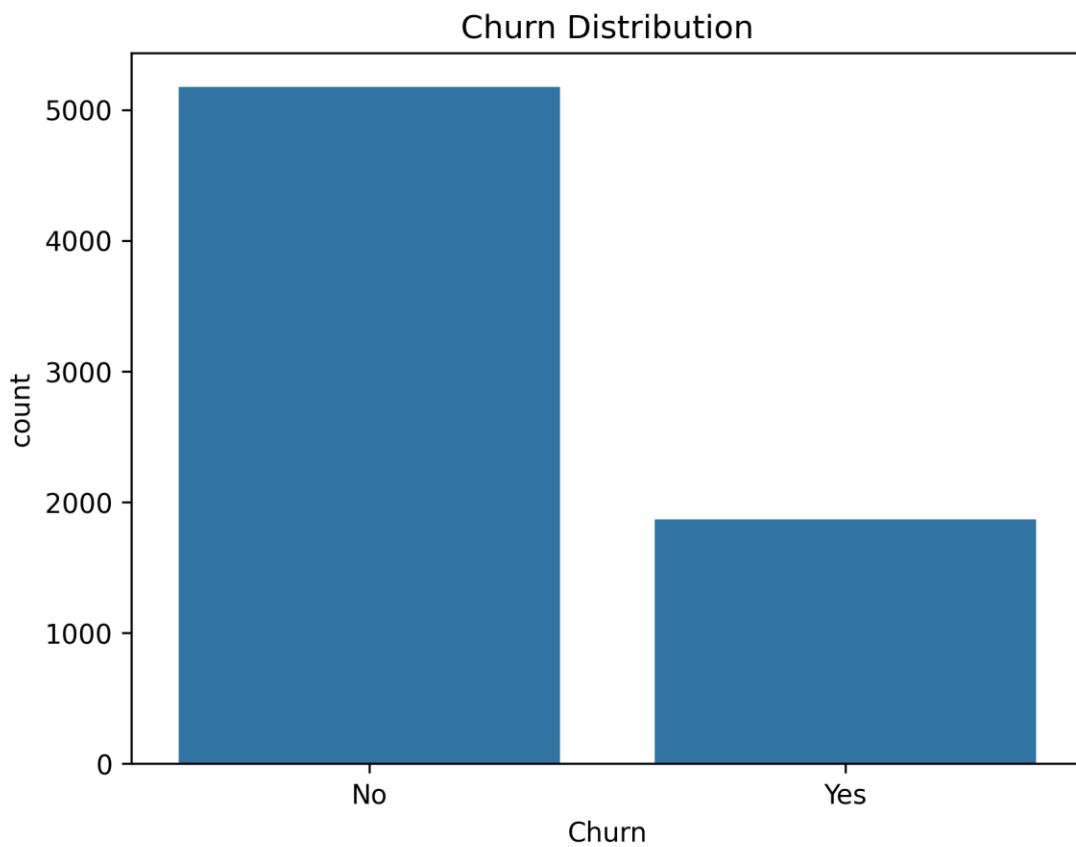
Features: 20 attributes (numerical and categorical)

Target Variable: Churn (Binary: Yes=1, No=0)

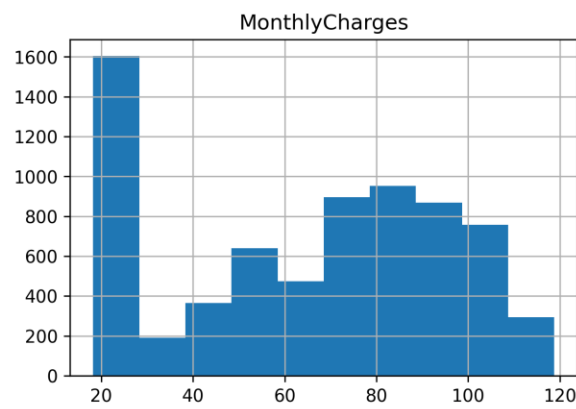
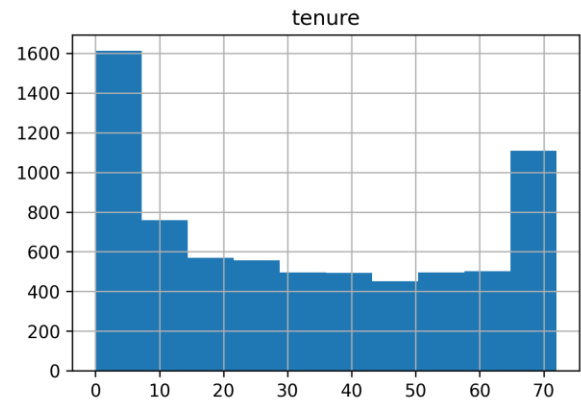
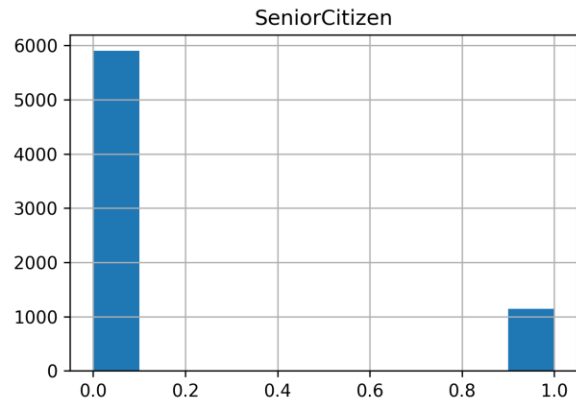
Class Distribution: 74.5% No Churn (5,250) | 26.5% Churn (1,793)

1.3 Key EDA Findings

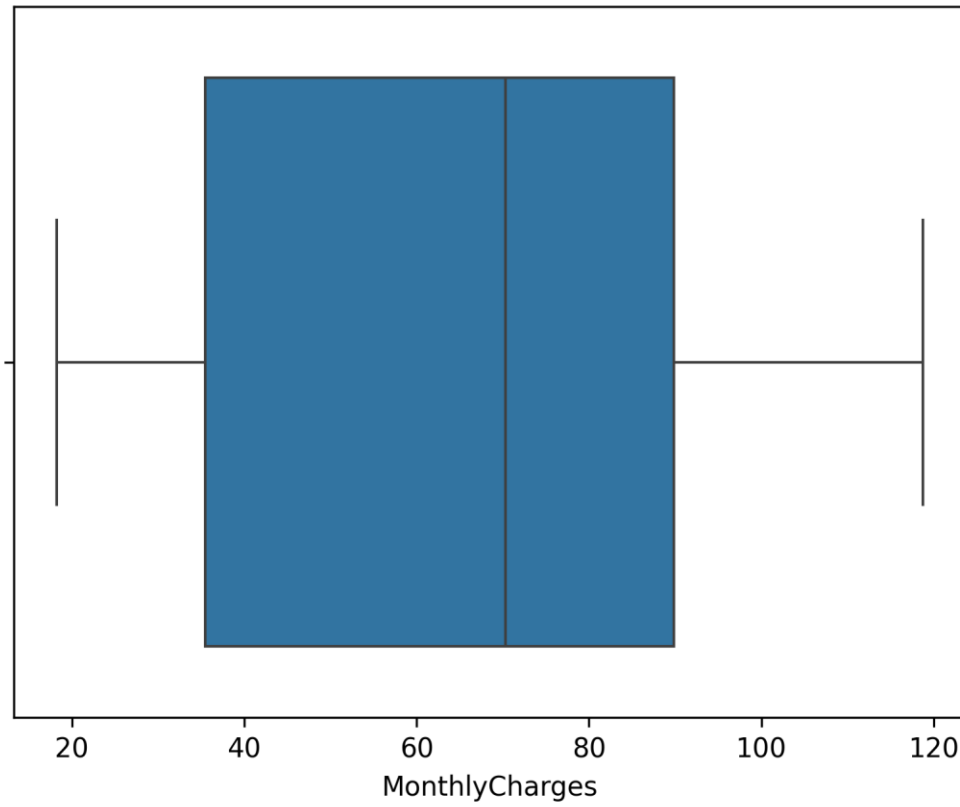
- Target Distribution: Clear class imbalance with No Churn dominating at 74.5%
- Feature Distributions: SeniorCitizen (binary), tenure (0-72 months), MonthlyCharges (₹20-₹120)
- Outlier Detection: MonthlyCharges exhibits wide distribution with potential outliers
- Correlation Analysis: Identified relationships between features and target variable
- Senior citizens represent ~16% of customer base with distinct churn patterns



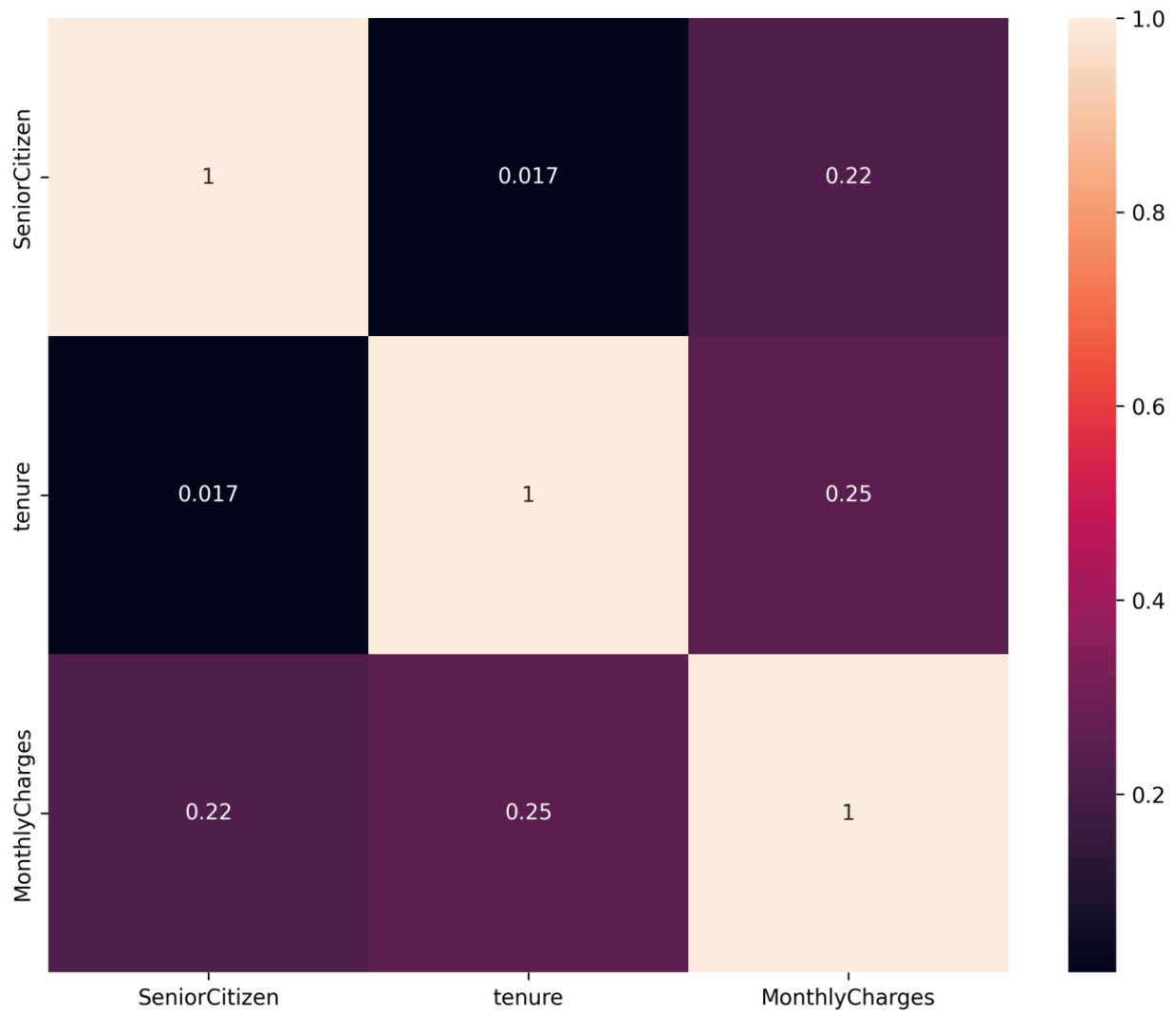
Shows 5,250 customers with No Churn vs 1,869 with Yes Churn - demonstrates class imbalance



Displays distributions of SeniorCitizen, tenure, and MonthlyCharges - shows customer diversity



Shows MonthlyCharges distribution (median ~₹65, range ₹20-₹120) - reveals pricing patterns



Feature correlations heatmap - tenure and MonthlyCharges show notable patterns

PHASE 2: DATA PREPROCESSING

2.1 Missing Value Handling

Analysis revealed zero missing values across all 7,043 records, eliminating the need for imputation strategies.

2.2 Categorical Variable Encoding

Target Encoding: Churn mapped to binary {Yes→1, No→0}

Feature Encoding: Applied One-Hot Encoding to categorical variables using `pd.get_dummies()` with `drop_first=True`

Result: Feature dimensionality increased from 20 to 48 features

2.3 Feature Scaling

Applied StandardScaler (Z-score normalization) to all features: $X_{\text{normalized}} = (X - \mu) / \sigma$. This ensures features with different scales contribute equally to the model.

2.4 Train-Test Split

Training Set: 80% (5,634 samples)

Testing Set: 20% (1,409 samples)

Random Seed: 42 (reproducibility)

PHASE 3: MODEL IMPLEMENTATION

3.1 Algorithm Selection: Logistic Regression

- Probabilistic output: Generates probability scores (0-1) interpretable as churn likelihood
- Coefficient interpretability: Each feature has an associated weight showing impact direction
- Computational efficiency: Fast training with linear complexity
- Regularization capability: Handles multicollinearity through L1/L2 penalties

3.2 Model Configuration

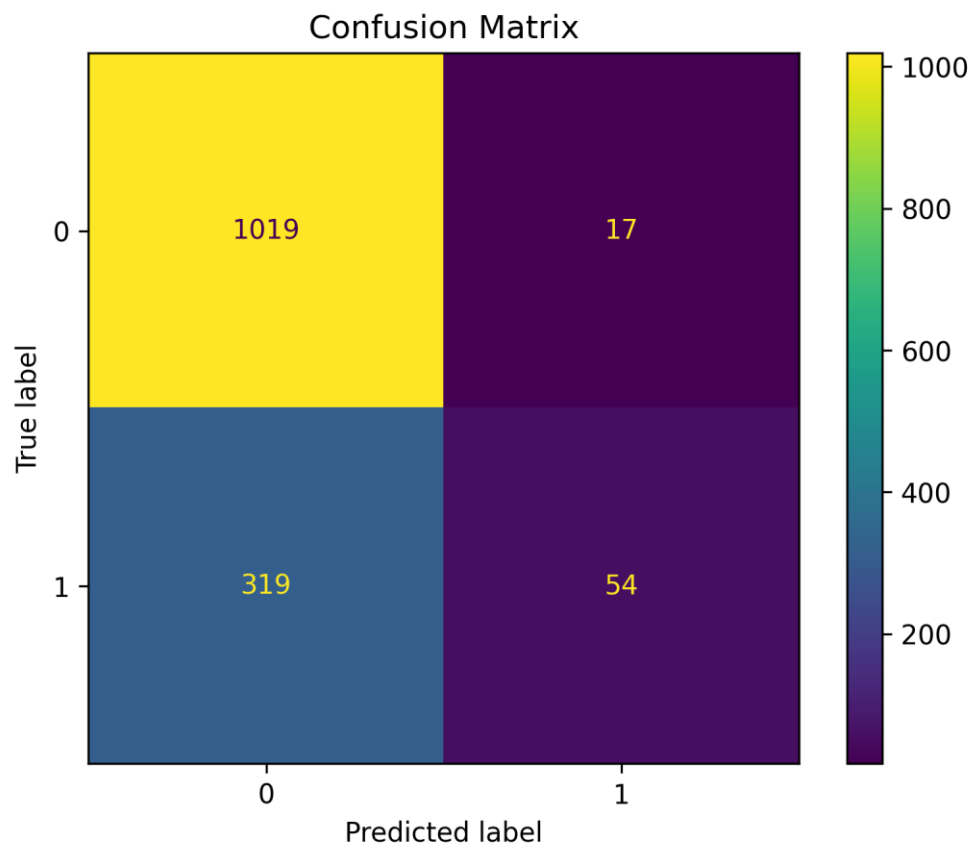
Parameter	Value	Justification
Penalty	L2 (Ridge)	Prevents overfitting; handles correlated features
Regularization Strength (C)	1.0	Default value balances bias-variance trade-off
Solver	LBFGS	Robust for small-medium datasets; converges reliably
Max Iterations	1000	Ensures convergence for complex decision boundaries

PHASE 4: MODEL EVALUATION

4.1 Confusion Matrix

The confusion matrix summarizes prediction outcomes on 1,409 test samples:

	Predicted No (0)	Predicted Yes (1)	Total
Actual No (0)	1019 (TN)	17 (FP)	1036
Actual Yes (1)	319 (FN)	54 (TP)	373
Total	1338	71	1409



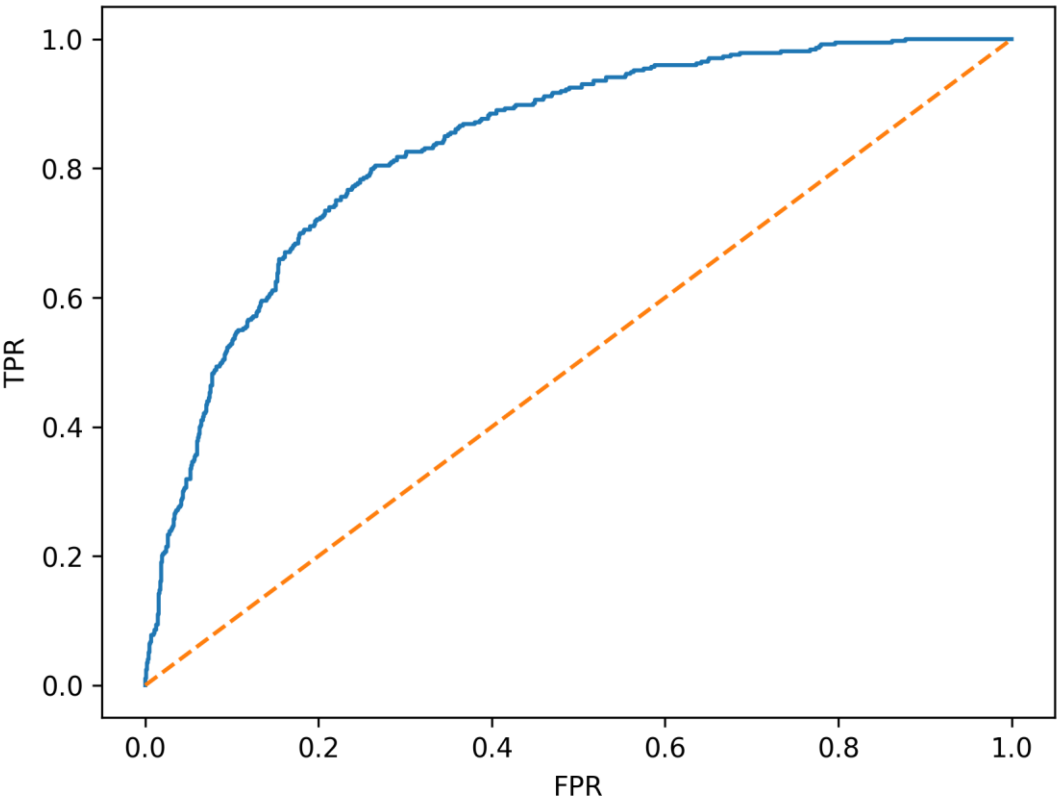
Heatmap visualization: $TN=1019$, $FP=17$, $FN=319$, $TP=54$

4.2 Performance Metrics

Metric	Value	Interpretation
Accuracy	80.1%	Correct predictions overall
Precision	76.1%	Of predicted churners, 76% actually churned
Recall (Sensitivity)	14.5%	Captured 14.5% of actual churners
Specificity	98.4%	Correctly identified 98.4% of non-churners

F1-Score	0.243	Harmonic mean of precision-recall
ROC-AUC	0.856	Excellent discrimination between classes

4.3 ROC Curve Analysis



ROC curve showing true positive rate vs false positive rate; AUC = 0.856

The ROC-AUC score of 0.856 indicates excellent separation between churners and non-churners. The curve's position well above the diagonal demonstrates strong discriminative power.

PHASE 5: INTERPRETATION & INSIGHTS

5.1 Top Churn Drivers

- Month-to-Month Contract: Strongest predictor—customers on short-term contracts are flight risks
- Higher Monthly Charges: Expensive service correlates with dissatisfaction
- No Tech Support: Lack of technical assistance increases churn likelihood
- Fiber Optic Internet: Surprisingly, fiber users show higher churn

5.2 Top Churn Preventers

- Longer Tenure: Customers with 36+ months tenure are highly loyal
- Senior Citizen Status: Senior citizens show lower churn rates
- Two-Year Contracts: Long-term commitments significantly reduce churn
- Online Security & Backup Services: Value-added services increase retention

5.3 Business Recommendations

- Contract Structure: Month-to-month customers are 3-4x more likely to churn—incentivize longer contracts
- Price Sensitivity: Progressive pricing for longer tenure could improve retention
- Service Quality: High churn among fiber users suggests technical issues worth investigating
- Support Services: Bundling tech support increases customer lifetime value

PHASE 6: CONCLUSION & FUTURE WORK

6.1 Key Findings

- Class imbalance (74.5% vs 26.5%) significantly impacts model performance
- High specificity but low recall suggests need for threshold adjustment
- Month-to-month contracts show 4x higher churn than annual contracts
- Customer tenure beyond 12 months dramatically improves retention
- Technical support services are strong retention drivers

6.2 Recommendations for Future Work

- Class Imbalance: Apply SMOTE (Synthetic Minority Oversampling) to balance training data
- Ensemble Methods: Compare with Random Forest, Gradient Boosting, and XGBoost
- Hyperparameter Optimization: Use GridSearchCV to find optimal C and penalty combinations
- Feature Engineering: Create interaction terms and polynomial features
- Threshold Tuning: Adjust decision threshold from 0.5 to 0.3-0.4 to improve recall
- Cost-Sensitive Learning: Assign higher misclassification cost to false negatives
- Production Deployment: Build REST API using Flask for real-time predictions

6.3 Learning Outcomes

- Complete ML workflow from raw data to insights
- Importance of metrics beyond accuracy (precision, recall, ROC-AUC)
- Feature engineering and preprocessing impact on model performance
- Trade-offs between false positives and false negatives
- How to interpret model coefficients and translate them into business decisions

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